Topic: Queue using Singly Linked List

Q. Describe the implementation of a Queue using a Singly Linked List. Detail the logic for the Enqueue and Dequeue operations.

> **Definition to Remember**

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Queue Implementation using Singly Linked List

To achieve O(1) time complexity for both Enqueue and Dequeue, we must maintain references to both ends of the list.

| Queue Pointer | Linked List Equivalent | Used For |

| :--- | :

2. Enqueue Operation (Insert at End)

Adds a new element to the rear of the queue.

Algorithm:

```text

### 3. Dequeue Operation (Delete from Beginning)

Removes the element from the front of the queue.

**Algorithm:**

```text

### 4. Advantages of Linked List Implementation

- **Dynamic Sizing:** Queue can grow infinitely (bounded only by heap memory). No "Queue Full" fixed limit.

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> **Must-Write Points (for 10 marks)**

> 1. Q

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> **Quick Recall**

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